

Advanced Computer Networks – PCS223

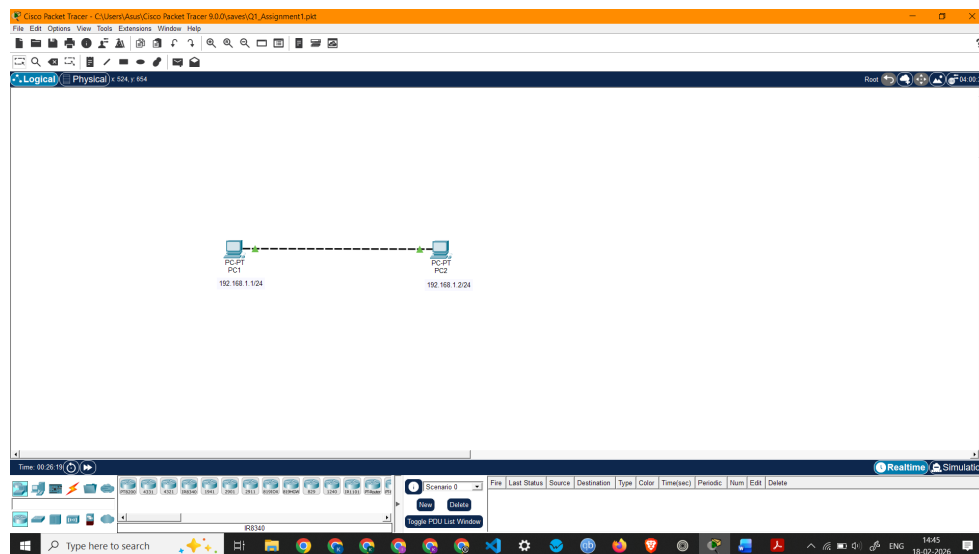
Assignment 3

Name – Dhruv Garg

Subgroup – CS2

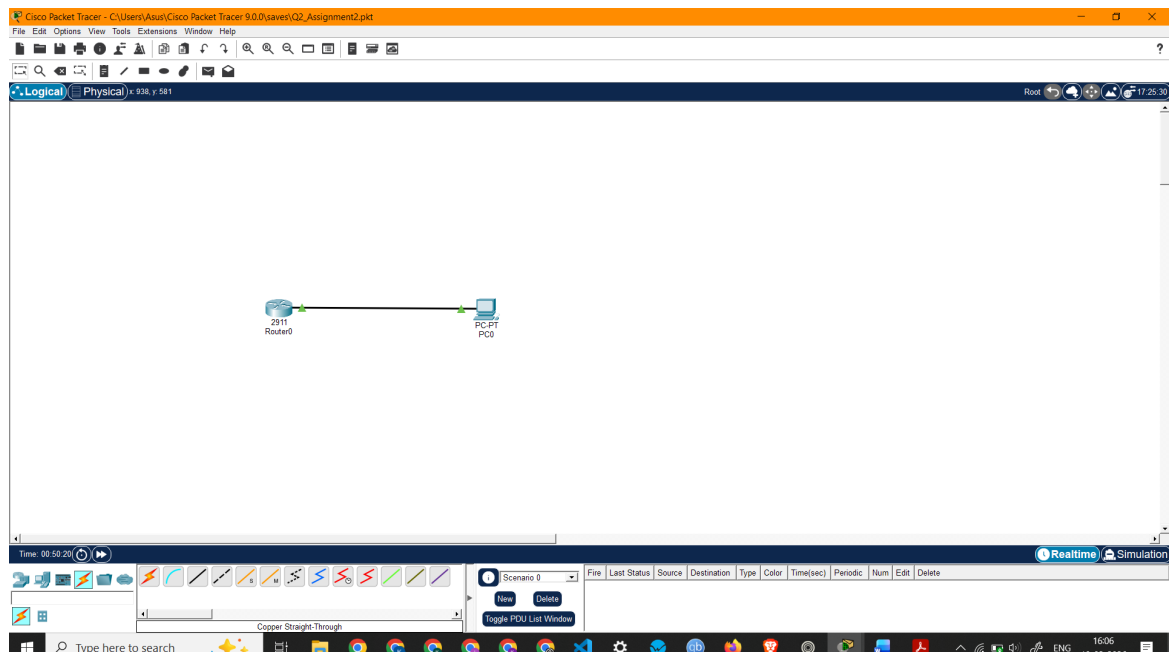
Roll No. – 8025320034

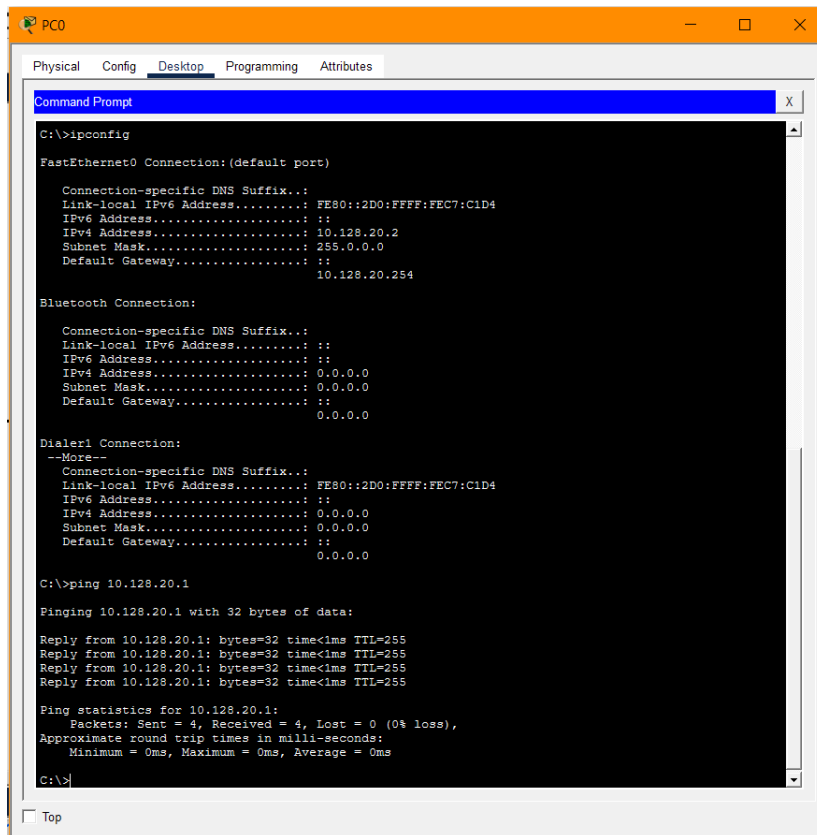
1. Connect one PC with another PC as per diagram below.



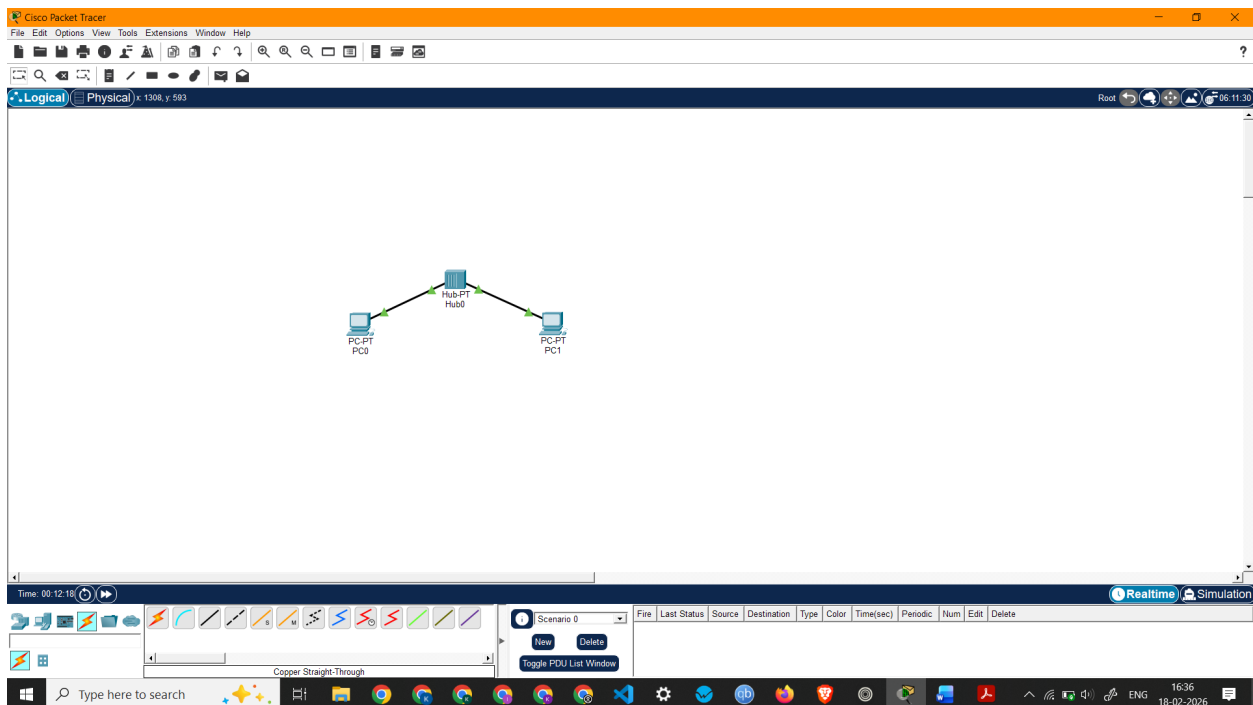
```
PC1
Physical Config Desktop Programming Attributes
Command Prompt
C:\>ping 192.168.1.2
Pinging 192.168.1.2 with 32 bytes of data:
Reply from 192.168.1.2: bytes=32 time=2ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms
C:\>
```

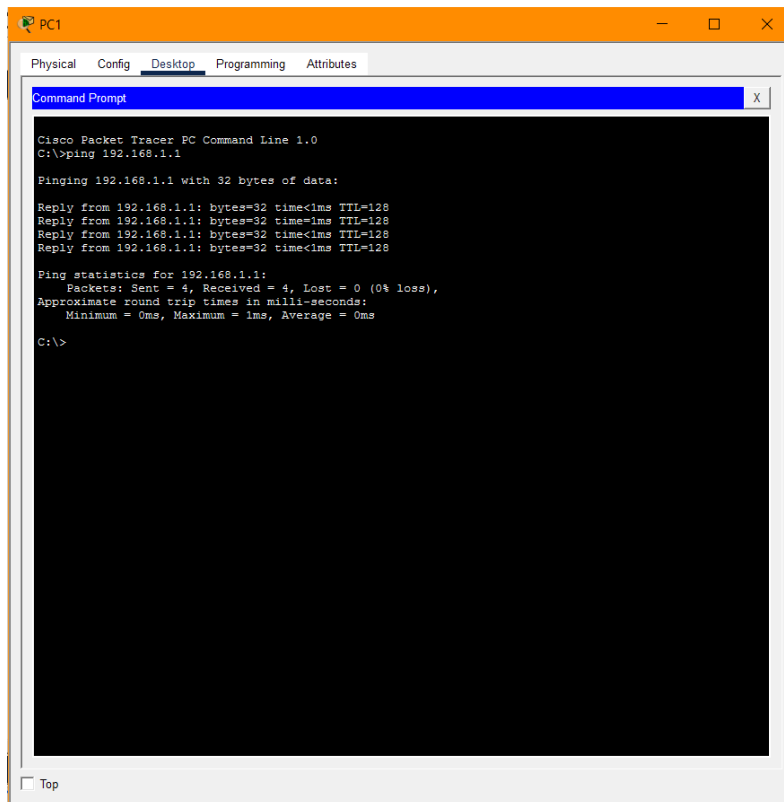
2. Connect a PC with Router Ethernet port as shown below.



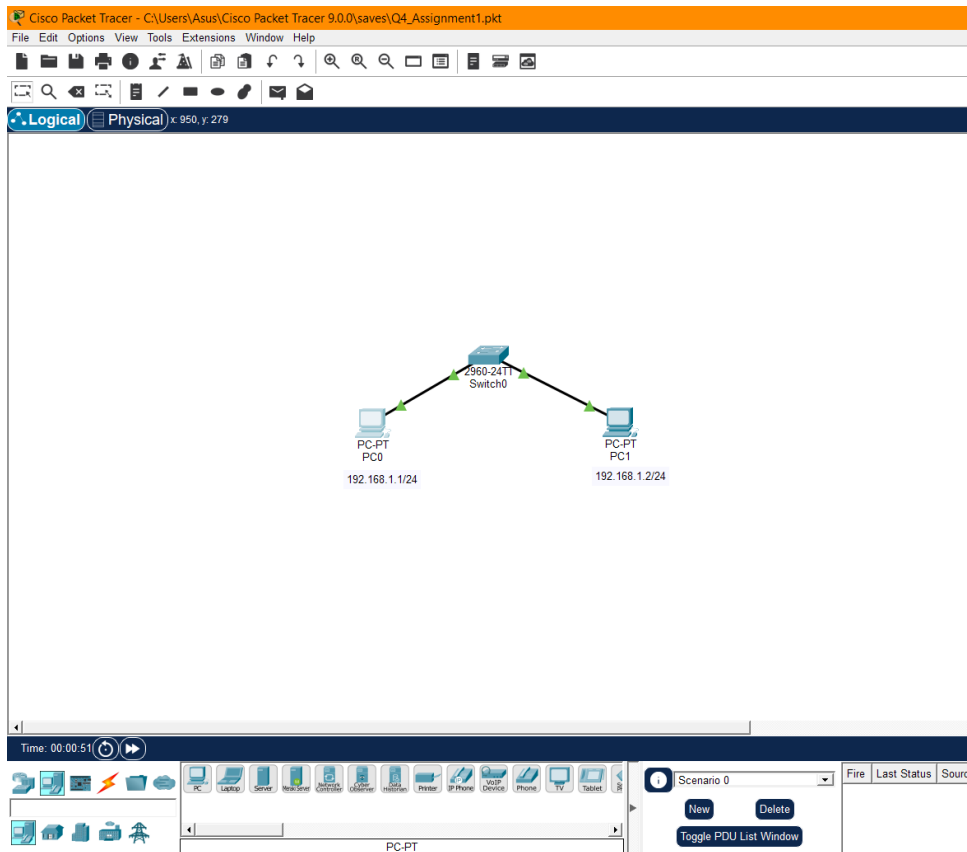


3. Create a network of 2 PCs as below.





4. Connect 2 PCs with a Switch below.



PC0

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\> ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

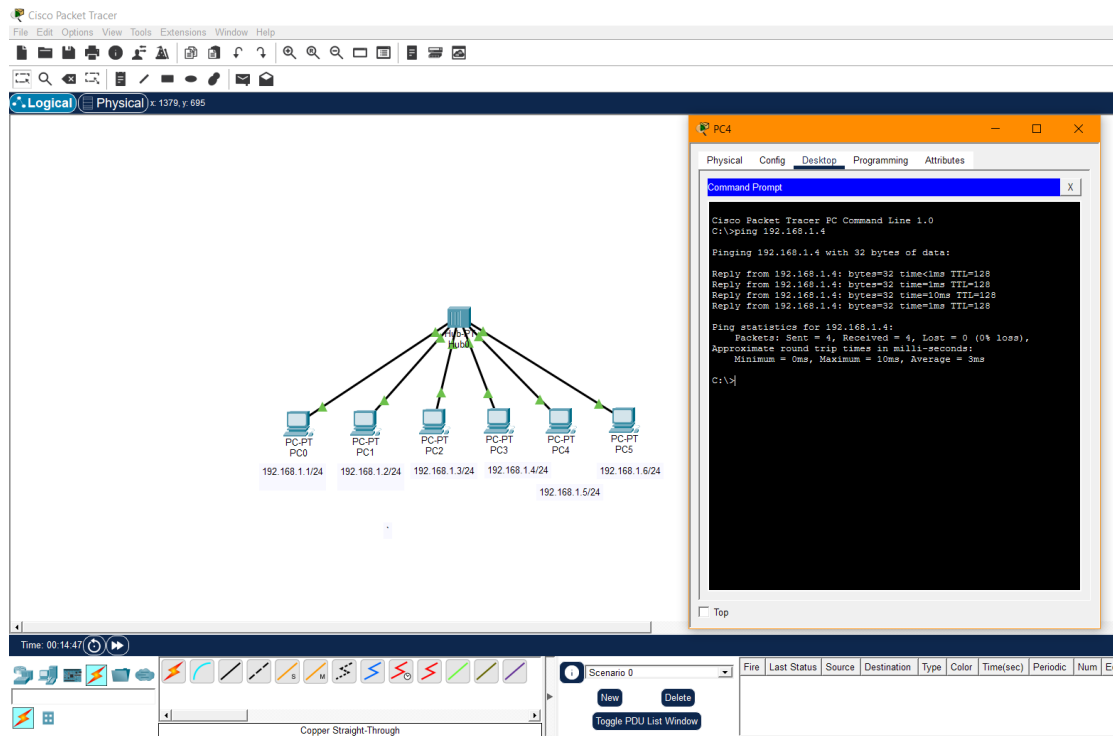
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

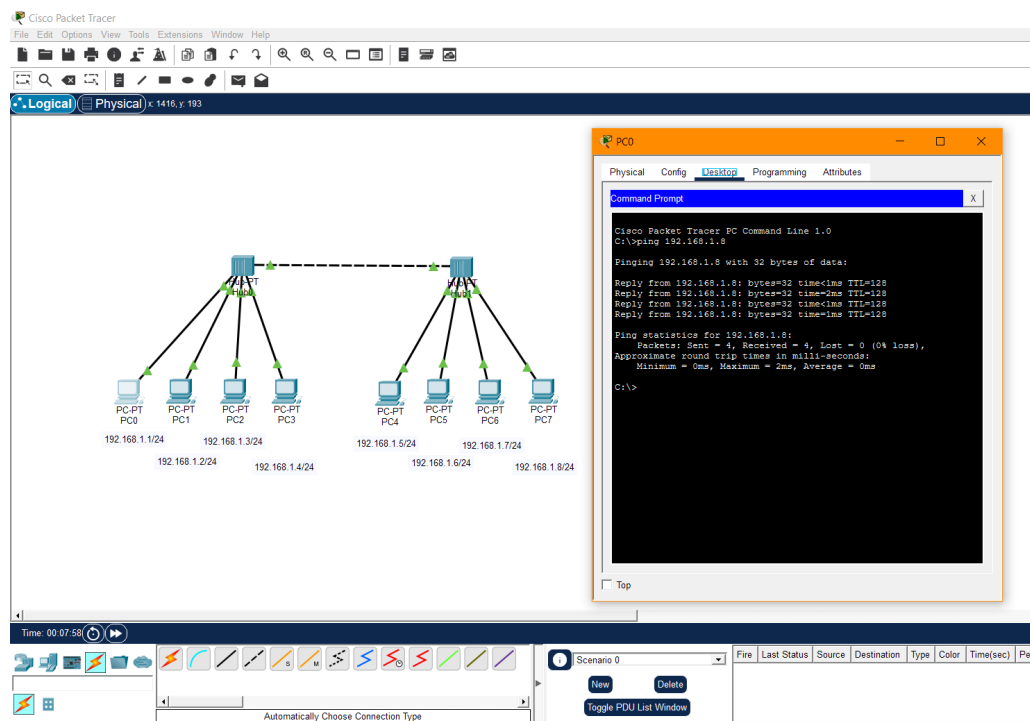
C:\>
```

Top

5. Create BUS topologies as below.

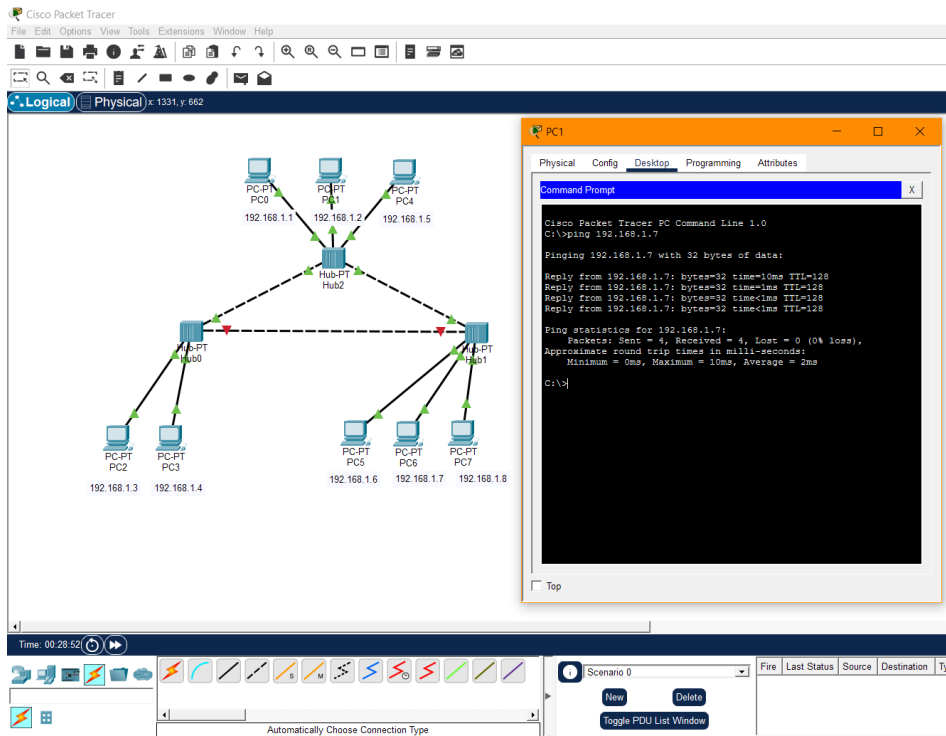


(a)

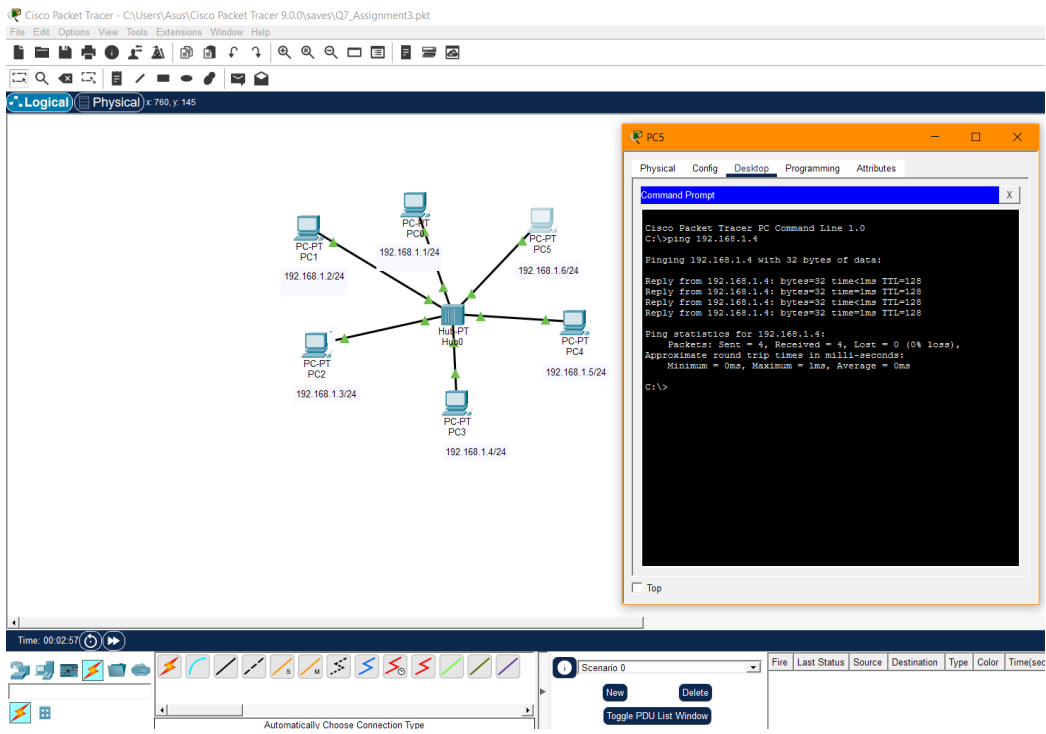


(b)

6. Create the following ring topology



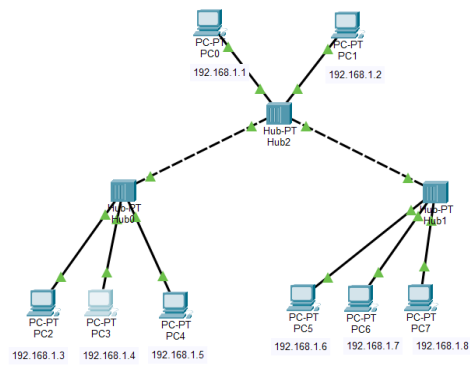
7. Create the following ring topology



8. Create the following ring topology



Logical Physical x 1370, y 626



PC3

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.7

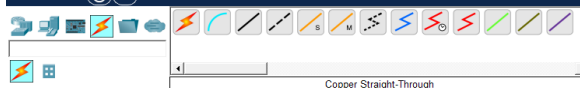
Pinging 192.168.1.7 with 32 bytes of data:
Reply from 192.168.1.7: bytes=32 time=10ms TTL=128
Reply from 192.168.1.7: bytes=32 time=1ms TTL=128
Reply from 192.168.1.7: bytes=32 time=10ms TTL=128
Reply from 192.168.1.7: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.1.7:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 10ms, Average = 5ms

C:\>
```

Top

Time: 00:03:48



Copper Straight-Through

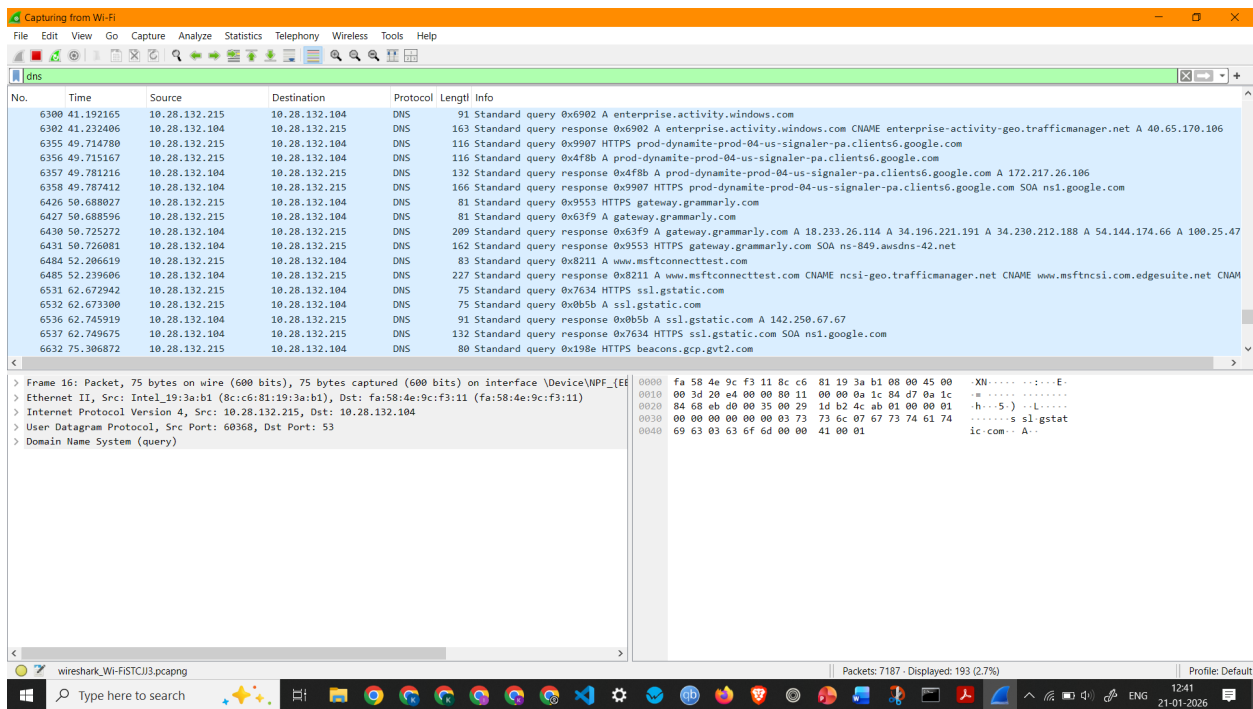
Scenario 0

New Delete

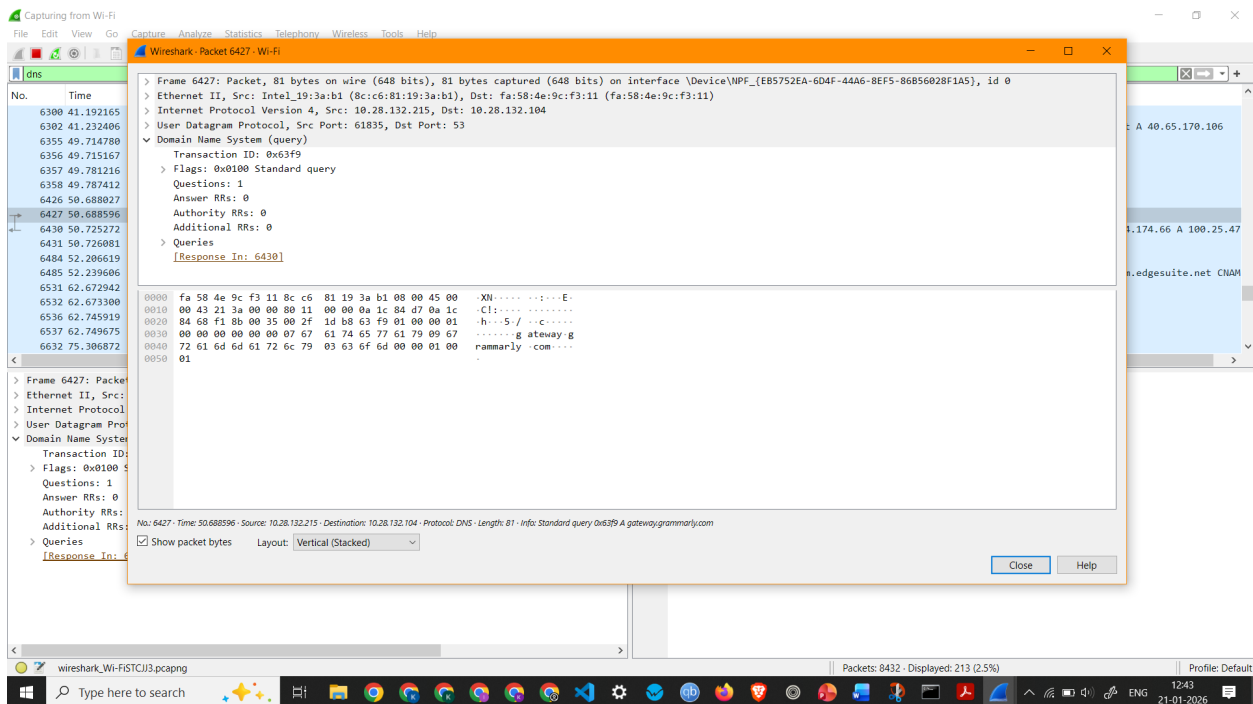
Toggle PDU List Window

Fire	Last Status	Source	Destination
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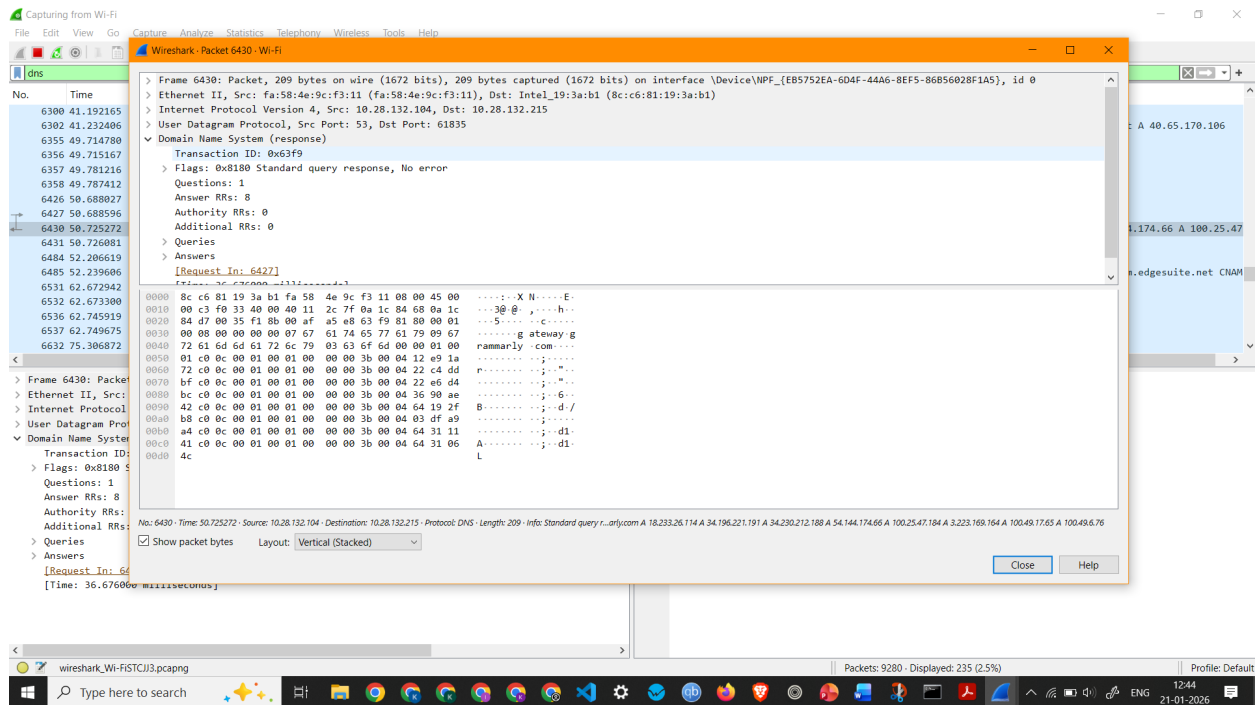
4. To capture and analyze DNS query and response packets



Screenshot showing DNS Packets Capture



Screenshot showing details of a DNS Packet 6427 (Standard query request)



Screenshot showing details of a DNS Packet 6430 (Standard query response)

Observation: DNS Packet 6427 –

- Interface Used: Wi-Fi
- Protocols Observed: Ethernet II, Internet Protocol Version 4, and User Datagram Protocol
- Source IP Address: 10.28.132.215
- Destination IP Address: 10.28.132.104
- Packet Length: 81 Bytes
- Queries - gateway.grammarly.com: type A, class IN

Observation: DNS Packet 6430 –

- Interface Used: Wi-Fi

- Protocols Observed: Ethernet II, Internet Protocol Version 4, and User Datagram Protocol
- Source IP Address: 10.28.132.104
- Destination IP Address: 10.28.132.215
- Packet Length: 209 Bytes

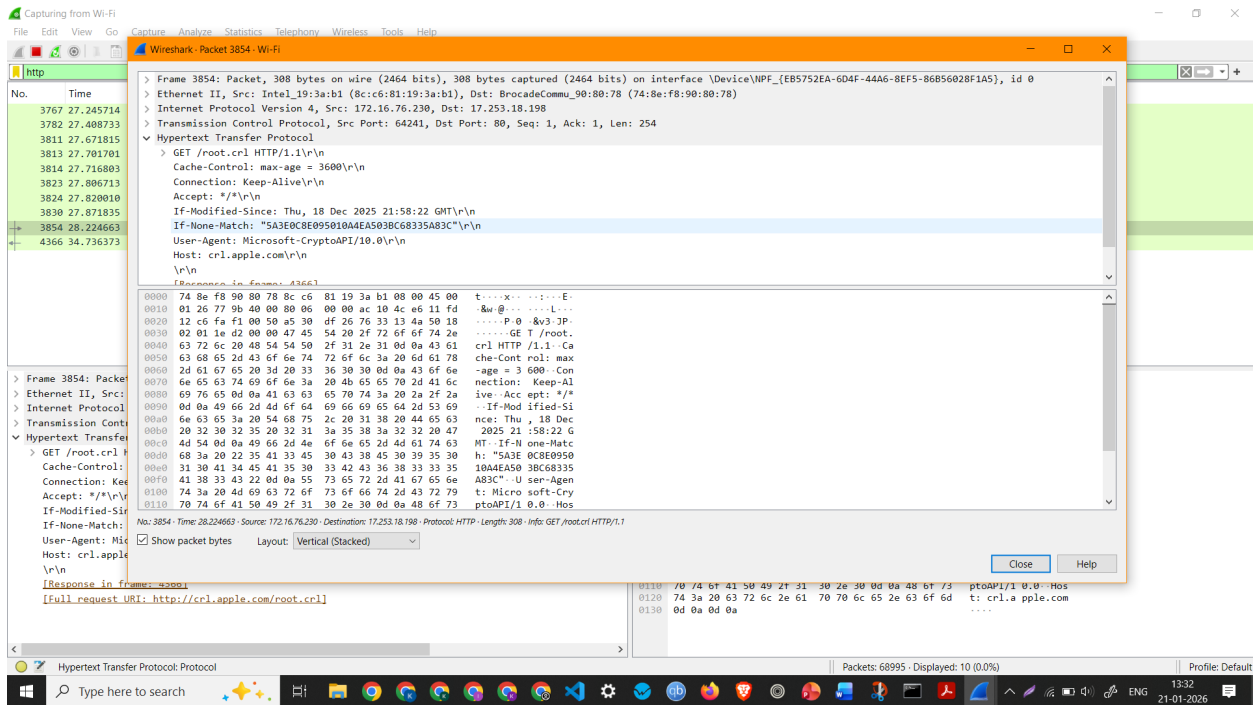
5. To capture and analyze HTTP packets

The screenshot displays the Wireshark network protocol analyzer interface. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. The main window is divided into three panes:

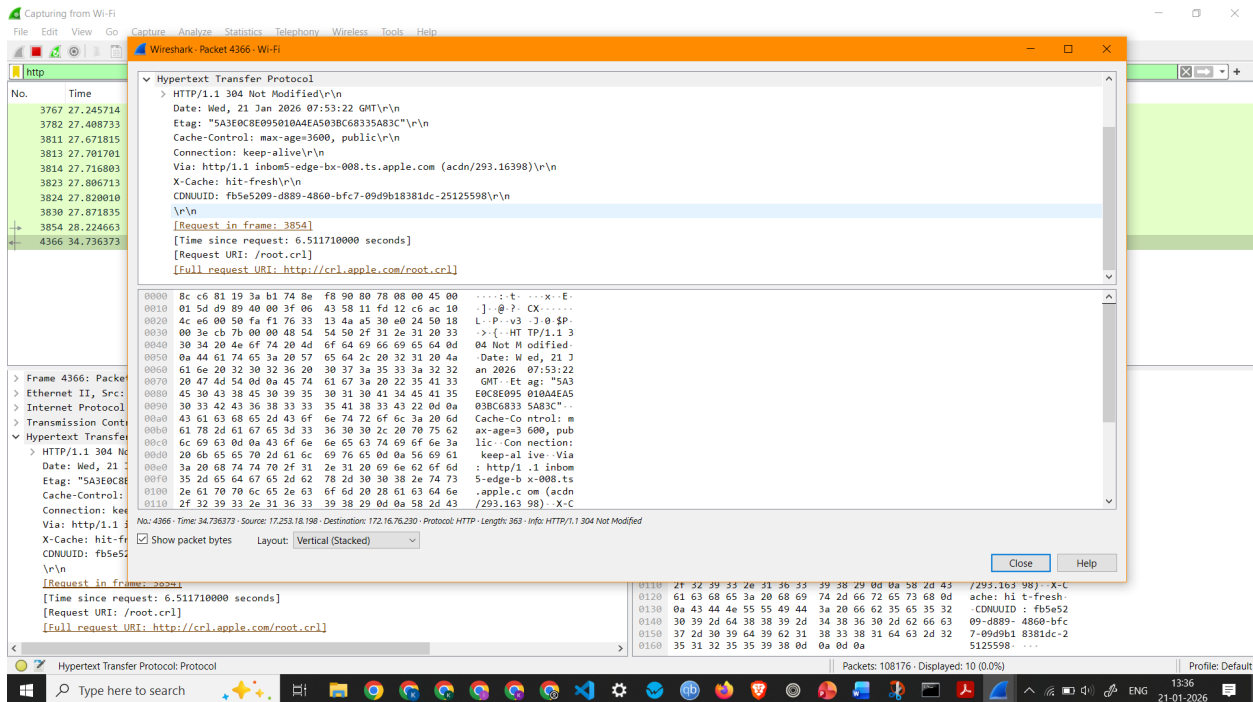
- Packet List:** Shows a list of captured packets. The selected packet is No. 3767, which is an HTTP GET request from 172.16.76.230 to 142.250.182.35.
- Packet Details:** Shows the hierarchical structure of the selected packet. The layers are:
 - Ethernet II, Src: Intel_19:3a:b1 (8c:c6:81:19:3a:b1), Dst: BrocadeComm_90:80:78 (74:8e:f8:90:80:78)
 - Internet Protocol Version 4, Src: 172.16.76.230, Dst: 142.250.182.35
 - Transmission Control Protocol, Src Port: 64239, Dst Port: 80, Seq: 1, Ack: 1, Len: 200
 - Hypertext Transfer Protocol
- Packet Bytes:** Shows the raw data of the selected packet in hexadecimal and ASCII. The ASCII column displays the decoded HTTP request, including the GET method, the requested resource (r/r4.cr1 HTTP/1.1), and various headers like User-Agent, Host, and Accept.

The bottom status bar indicates that 15511 packets have been captured and 10 (0.1%) are displayed. The system clock shows 13:25 on 21-01-2026.

Screenshot showing HTTP Packets Capture



Screenshot showing details of a HTTP Packet (HTTP Request)



Screenshot showing details of a HTTP Packet (HTTP Response)

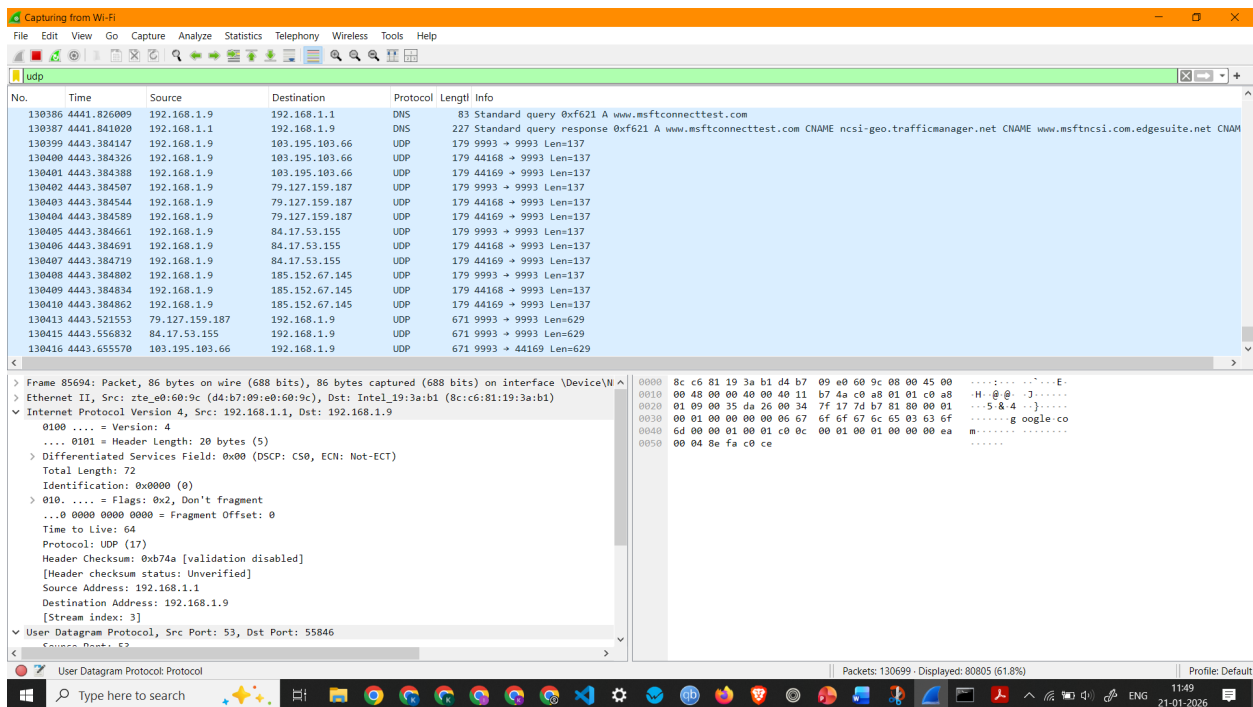
Observation: HTTP Request 3854 –

- Interface Used: Wi-Fi
- Protocols Observed: Ethernet II, Internet Protocol Version 4, Transmission Control Protocol, Hypertext Transfer Protocol
- Source IP Address: 172.16.76.230
- Destination IP Address: 17.253.18.198
- Packet Length: 308 Bytes

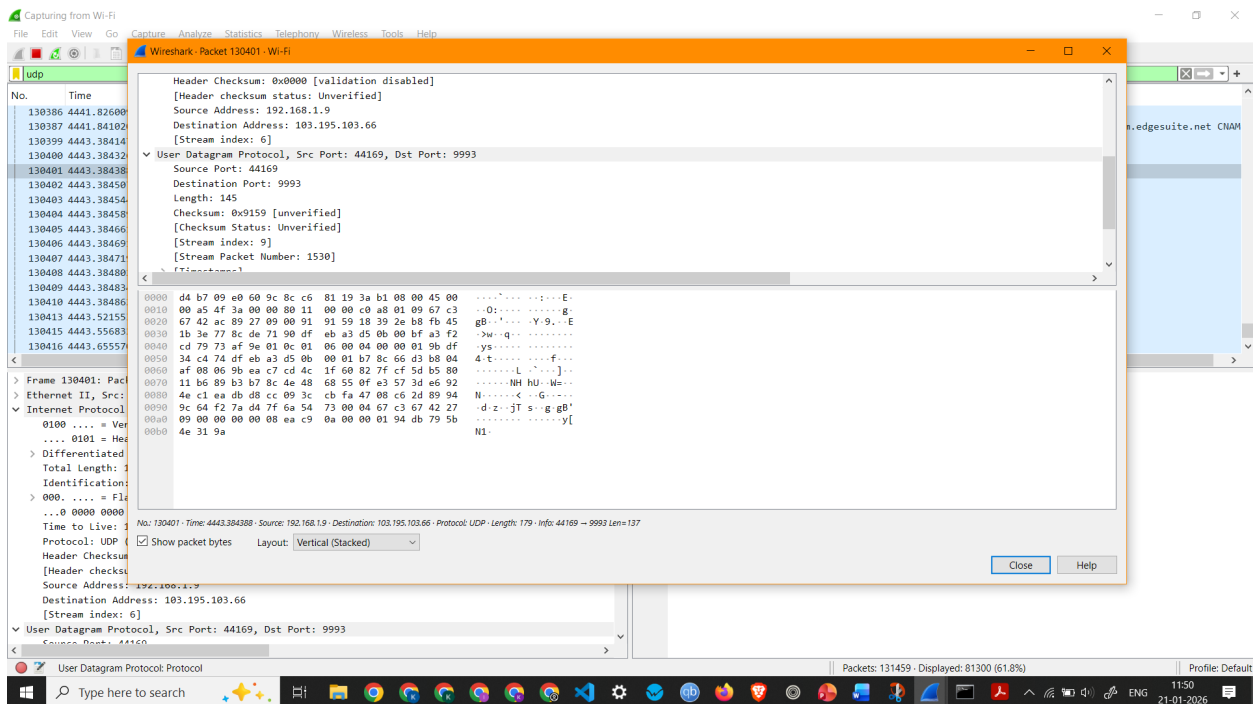
Observation: HTTP Response 4366 –

- Interface Used: Wi-Fi
- Protocols Observed: Ethernet II, Internet Protocol Version 4, Transmission Control Protocol, Hypertext Transfer Protocol
- Source IP Address: 17.253.18.198
- Destination IP Address: 172.16.76.230
- Packet Length: 363 Bytes

3. To capture and analyze UDP packets



Screenshot showing UDP Packets Capture



Screenshot showing details of a UDP Packet

Observation: UDP Packet 130401 –

- Interface Used: Wi-Fi
- Protocols Observed: Ethernet II, Internet Protocol Version 4, and User Datagram Protocol
- Source IP Address: 192.168.1.9
- Destination IP Address: 103.195.103.66
- Packet Length: 179 Bytes

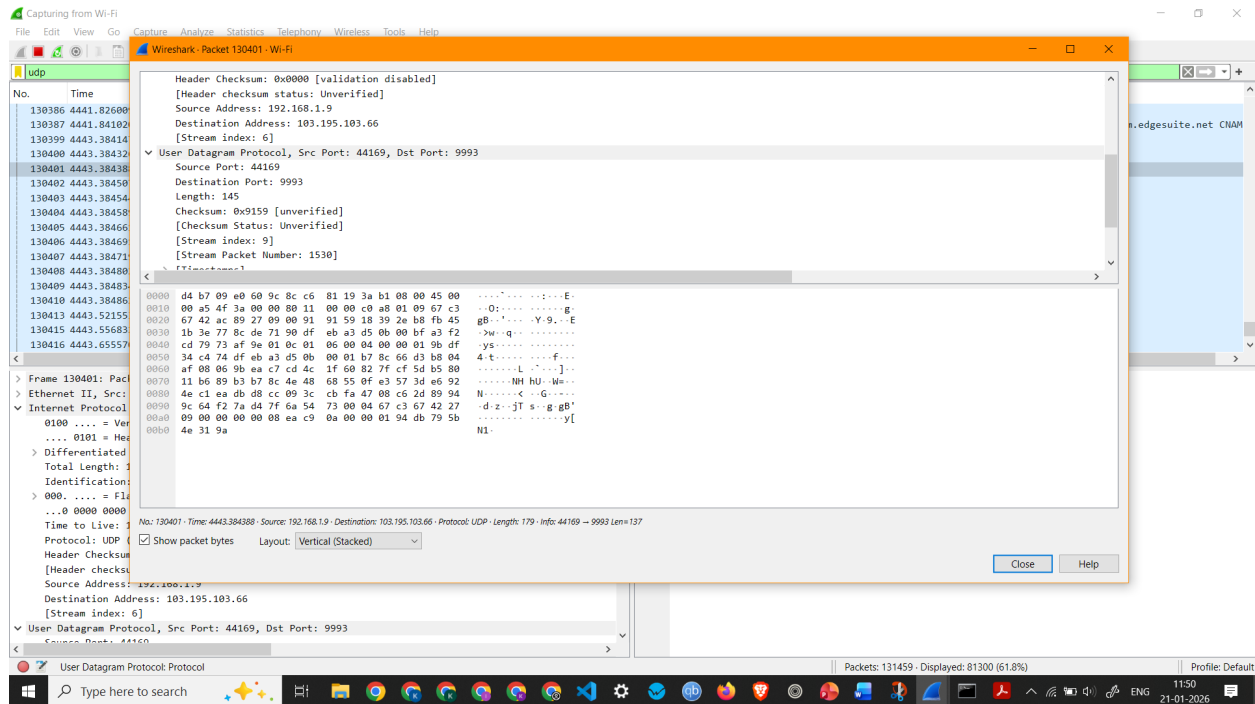
3. To capture and analyze UDP packets

The screenshot displays the Wireshark network traffic analysis tool. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. The main window is divided into three panes:

- Packets List:** A table showing a list of captured packets. Packet 130401 is highlighted, showing it is a UDP packet from source 192.168.1.9 to destination 103.195.103.66, with a length of 179 bytes.
- Packet Details:** A tree view showing the protocol stack for packet 130401: Ethernet II, Internet Protocol Version 4, and User Datagram Protocol. The UDP section shows the source port as 53 and the destination port as 55846.
- Packet Bytes:** A hex dump and ASCII representation of the packet data. The hex dump shows the raw bytes of the packet, and the ASCII representation shows the corresponding text, including the word "google" in the payload.

The bottom status bar indicates that 130699 packets were captured, with 80805 (61.8%) displayed. The system clock shows 11:49 on 21-01-2026.

Screenshot showing UDP Packets Capture



Screenshot showing details of a UDP Packet

Observation: UDP Packet 130401 –

- Interface Used: Wi-Fi
- Protocols Observed: Ethernet II, Internet Protocol Version 4, and User Datagram Protocol
- Source IP Address: 192.168.1.9
- Destination IP Address: 103.195.103.66
- Packet Length: 179 Bytes

3. To capture and analyze UDP packets

Observation: UDP Packet 130401 –

- Interface Used: Wi-Fi
- Protocols Observed: Ethernet II, Internet Protocol Version 4, and User Datagram Protocol
- Source IP Address: 192.168.1.9
- Destination IP Address: 103.195.103.66
- Packet Length: 179 Bytes

3. To capture and analyze UDP packets

The screenshot displays the Wireshark interface with a packet capture from a Wi-Fi interface. The packet list shows several UDP packets from 192.168.1.9 to 103.195.103.66. Packet 130401 is selected, and its details are shown in the lower pane.

Packet List:

No.	Time	Source	Destination	Protocol	Length	Info
130386	4441.826009	192.168.1.9	192.168.1.1	DNS	83	Standard query 0xf621 A www.msftconnecttest.com
130387	4441.841020	192.168.1.1	192.168.1.9	DNS	227	Standard query response 0xf621 A www.msftconnecttest.com CNAME ncsi-geo.trafficmanager.net CNAME www.msftncsi.com.edgesuite.net CNAM
130399	4443.384147	192.168.1.9	103.195.103.66	UDP	179	9993 → 9993 Len=137
130400	4443.384326	192.168.1.9	103.195.103.66	UDP	179	44168 → 9993 Len=137
130401	4443.384388	192.168.1.9	103.195.103.66	UDP	179	44169 → 9993 Len=137
130402	4443.384507	192.168.1.9	79.127.159.187	UDP	179	9993 → 9993 Len=137
130403	4443.384544	192.168.1.9	79.127.159.187	UDP	179	44168 → 9993 Len=137
130404	4443.384589	192.168.1.9	79.127.159.187	UDP	179	44169 → 9993 Len=137
130405	4443.384661	192.168.1.9	84.17.53.155	UDP	179	9993 → 9993 Len=137
130406	4443.384691	192.168.1.9	84.17.53.155	UDP	179	44168 → 9993 Len=137
130407	4443.384719	192.168.1.9	84.17.53.155	UDP	179	44169 → 9993 Len=137
130408	4443.384802	192.168.1.9	185.152.67.145	UDP	179	9993 → 9993 Len=137
130409	4443.384834	192.168.1.9	185.152.67.145	UDP	179	44168 → 9993 Len=137
130410	4443.384862	192.168.1.9	185.152.67.145	UDP	179	44169 → 9993 Len=137
130413	4443.521553	79.127.159.187	192.168.1.9	UDP	671	9993 → 9993 Len=629
130415	4443.556832	84.17.53.155	192.168.1.9	UDP	671	9993 → 9993 Len=629
130416	4443.655570	103.195.103.66	192.168.1.9	UDP	671	9993 → 44169 Len=629

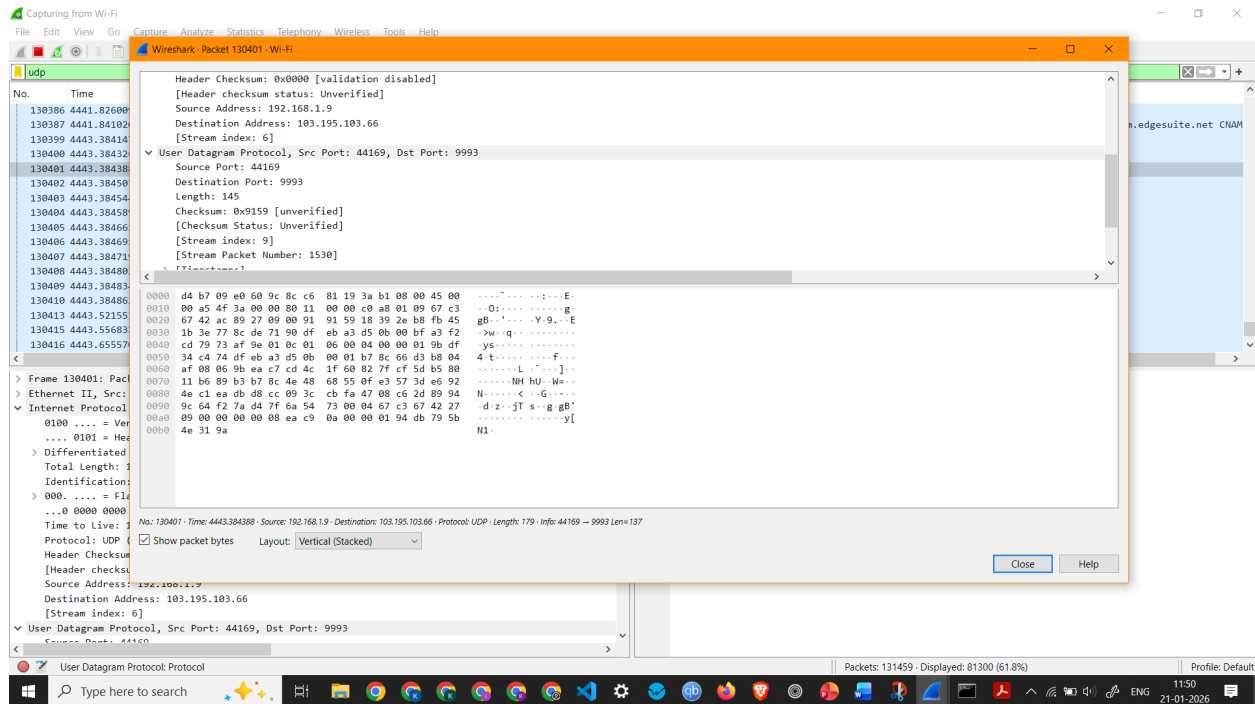
Packet 130401 Details:

- Frame 85694: Packet, 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface \\Device\NPF...
- Ethernet II, Src: zte_e0:60:9c (d4:b7:09:e0:60:9c), Dst: Intel_19:3a:b1 (8c:c6:81:19:3a:b1)
- Internet Protocol Version 4, Src: 192.168.1.1, Dst: 192.168.1.9
 - 0100 = Version: 4
 - 0101 = Header Length: 20 bytes (5)
 - > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
 - Total Length: 72
 - Identification: 0x0000 (0)
 - > 010. = Flags: 0x2, Don't fragment
 - ...0 0000 0000 0000 = Fragment Offset: 0
 - Time to Live: 64
 - Protocol: UDP (17)
 - Header Checksum: 0xb74a [validation disabled]
 - [Header checksum status: Unverified]
 - Source Address: 192.168.1.1
 - Destination Address: 192.168.1.9
 - [Stream index: 3]
- User Datagram Protocol, Src Port: 53, Dst Port: 55846

Packet Bytes:

Offset	Hex	ASCII
0000	8c c6 81 19 3a b1 d4 b7 09 e0 60 9c 08 00 45 00	E-
0010	00 48 00 00 40 00 40 11 b7 4a c0 a8 01 01 c0 a8	..H..@..:.....
0020	01 09 00 35 da 26 00 34 7f 17 7d b7 01 00 00 01	...5&4...}
0030	00 01 00 00 00 00 06 67 6f 6f 67 6c 65 03 63 6f	g oogle co
0040	6d 00 00 01 00 01 c0 0c 00 01 00 01 00 00 00 ea	m.....
0050	00 04 8e fa c0 ce	

Screenshot showing UDP Packets Capture



Screenshot showing details of a UDP Packet

Observation: UDP Packet 130401 –

- Interface Used: Wi-Fi
- Protocols Observed: Ethernet II, Internet Protocol Version 4, and User Datagram Protocol
- Source IP Address: 192.168.1.9
- Destination IP Address: 103.195.103.66
- Packet Length: 179 Bytes